

Constructing CFTs from Category Data

**Algorithms, lattice Virasoro generators,
operator-algebraic renormalisation, and local fusion-space models**

cft-anyons research collaboration

Research synthesis with Astra cross-refereeing

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The objective. Discover an algorithm, or functor, that takes fusion or modular tensor category data to concrete lattice systems and microscopic Virasoro generators, with an OAR limit through Wilson's triangle that is a rigorous CFT with the starting modular data.

What has been achieved. An exact-input Ising compiler and declared tensor-product branch; explicit lattice fermion and coset stress generators; quantitative state, mode, product, resolvent and unitary convergence results; a positive constrained finite-algebra OAR construction; reproducible exact and numerical witnesses; and independent skeptical cross-reviews.

What has not been achieved. There is no general category-to-CFT functor. The new coset descendant construction has not been identified with a categorical fusion-space basis on which the Hamiltonian acts locally. The tricritical modular-input compiler was proposed but not implemented.

Locality, as clarified by Tobias. Global admissibility constraints may be fusion constraints built into the Hilbert space. They do not by themselves make a model nonlocal. The continuing target is a local model of fusion spaces, with those constraints incorporated automatically.

This report separates existing source theorems, local structured proofs, finite computational checks and unresolved proposals. Proofs use Lamport notation; none is represented as proof-assistant certified. All underlying sources, code, run bundles and referee reports are retained in the repository.

Executive synthesis

The strongest completed input-to-output branch begins with the standard chiral Ising modular datum, or an explicit product of such data. Exact cyclotomic arithmetic checks the input. The output contains actual Majorana lattice Hamiltonians, ground states, parity-compatible CAR refinements and corrected Koo–Saleur modes. A local reconstruction argument identifies the NS even-Fermi interval net, and the sourced Ising-net sector theorem supplies the complete modular data. This branch is a supported construction, not evidence that arbitrary categorical input is realizable by the same code.

The main new alternative is the coset route. Finite flavored CAR currents give explicit quartic Sugawara-difference stress polynomials. Their high-current tail vanishes exactly on each fixed free-energy subspace; sufficiently large microscopic windows reproduce each fixed mode and every fixed finite word. Coordinated Fourier and current cutoffs then give smooth-stress core, strong-resolvent and unitary convergence using a sourced energy bound.

There is also a decisive negative result: the raw finite quartic zero mode is not positive. An explicit polarized state has expectation $-(M - R)(M - R - 1)/12$. Extracting finite vacuum-descendant spaces and adding a positive complementary energy gives a different positive model. An independent scalar algebra summand supplies genuine unital OAR maps and compatible dynamics. These operations are constructive, but their current presentation supplies no bounded-range fusion-space implementation.

The methodological results connect these branches. Summable Wilson-row drift gives an effective state construction; graph-norm estimates control products of unbounded generators; finite-group averaging transports invariant states and represented algebras. Referees caught substantive mistakes and missing hypotheses, including the difference between generator compatibility and Wilson-state convergence, total versus factorwise parity, and the need to distinguish a norm inductive limit from represented field limits.

The final direction is therefore specific: preserve the constructive OAR and symmetry machinery, and find a categorical fusion-space description in which the required Hamiltonians and refinement operations are local. Replacing that task by arbitrary global projection would not complete the programme.

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1 The programme, its input, and the inherited starting point

1.1 What a completed machine must produce

The required output is more than a Hamiltonian with a plausible low-energy spectrum. It must connect categorical input to a finite microscopic family, its observable algebras, states and refinements, explicit symmetry generators, a mathematically controlled OAR limit, and the modular data of the resulting local CFT. The category-to-model construction may make choices, but those choices must be visible. If it is called a functor, the objects, morphisms, composition law and transported physical data must also be specified.

The working output tuple is

$$((A_N, H_N, \omega_N), \alpha_{N'}^N, L_{N,n}, (\mathcal{A}, \mathcal{H}, \Omega, U), \text{modular comparison}).$$

Here the first entries are microscopic; the last two require analysis and sector theory. The present code supplies this tuple on its declared Ising domain. The later constrained coset construction supplies a different finite-algebra OAR prescription with a local continuum reconstruction, while its local fusion-space implementation remains open.

1.2 Category, modular data, and additional conformal information

A fusion category alone does not specify modular S, T matrices. A braided target, or a distinct defect/soliton target, must be declared. For modular input the minimum fidelity requirement here is matching modular data under a unit-preserving relabelling. A braided equivalence is a stronger assertion: the registered Mignard–Schauenburg examples show that modular data do not determine a modular category up to braided equivalence.

The twist diagonal $\Theta = \text{diag}(\theta_a)$ and the character matrix $T = e^{-2\pi ic/24}\Theta$ are kept distinct. A real central-charge lift is not silently read off from topological data. Category data can admit multiple conformal realizations; tensoring with a holomorphic net preserves the representation category while changing conformal information. This does not rule out an algorithm that makes an explicit choice of realization.

Evidence. `cft_machine/research/realisation_and_fidelity.md`; Evans–Gannon source lines 250–260, 333–335, 473–480; Bischoff source lines 364–408; Mignard–Schauenburg source lines 121–129, 175–179, all under `references/cft-machine/realisation/`.

1.3 The inherited lab book

The existing project supplied several useful components and warnings. They were treated as inspiration and checked against the current source/code record, rather than adopted as an exclusive route.

Block	Role in the present investigation
CA-02–09	Many-particle Hilbert grammar, sectors, and Fibonacci fusion-path kinematics. The full categorical input includes coherence data beyond path counts.
CA-10–22	Lattice symmetry candidates, energy or mass first moments, and the distinction between Lorentzian and Galilean targets.
CA-23–33	Gaussian dispersion/current diagnostics, explicit normalization and boundary policies, and a distinction between symbol calculations and full representation/continuum theorems.

CA-34–61	Qubit local equations and moment-SDP filters. The recorded 99-point exact first-moment scan rejected every point for that route; this did not disprove the known scaling limits of its inputs.
CA-62–74	Algebra-first categorical construction, variable-number anyons, corner refinements, categorical residuals, and the finite Jones–Wenzl kernel calculation. This was the repository endpoint at the start of the new work.

The qubit scan’s local run reports 9 current collapses, 27 conservation-witness failures and 63 exact boost-witness failures. Its relaxed successor uses different, explicitly scoped verdicts. These results motivated avoiding an exact full-Hilbert-space lattice Virasoro demand.

The latest inherited Fibonacci result was the rank-322 dilute image at $L = 4$, with a one-dimensional Jones–Wenzl kernel in the 323-dimensional diagram algebra. Its surjectivity conclusion uses a numerical rank input. The initial orientation reran its 113 assertions; none of the new work expanded the Fibonacci computation beyond the existing $L \leq 4$ budget.

Evidence. Root `report/SHARD_CATALOG.md`;
`runs/2026-05-31-qubit-candidate-scan/summary.toml`;
`report/sections/74_jw_kernel_decision.tex`; `test/runtests.jl`, CA-74 testset. These are inherited results, not new continuum conclusions of this report.

1.4 Scope and proof status in this report

Source theorem. A result imported from a registered primary source, with its hypotheses checked for the stated application.

Local proof. A derivation given in the repository using Lamport’s ASSUME/PROVE, hierarchical steps and explicit dependencies.

Finite witness. A reproducible computation asserting a known value or invariant, with a producer, run bundle and regression checks.

Proposal. An unimplemented construction or a missing mathematical interface. It is not promoted by a favorable finite test.

The independent referee reports assess particular claims under particular hypotheses. They are evidence of scrutiny, not formal proof certificates. Every distinction becomes important below: a strong operator limit need not be a norm inductive limit; a finite stress difference need not be a positive Hamiltonian; and a globally constrained space need not already be a categorical fusion space with local operators.

2 The CFT Machine and Its Quantitative OAR Contract

2.1 Sources, Dependencies, and Status

This shard records Tobias’s 2026-09-22 formulation of the programme and the new workspace `cft_machine/`. It does not assert a universal construction. Dependencies: CA-62, CA-67, CA-72, CA-73; CONVENTIONS (u)–(y).

2.2 The Actual Construction Target

Programme. Discover an algorithm, or functor, from categorical data to concrete lattice systems and microscopic Virasoro generators, whose limit through operator-algebraic renormalisation (OAR) and Wilson’s triangle is a rigorous CFT with the input modular data. Indeterminate-number anyon spaces are one mechanism; they do not restrict the search.

A successful output names the microscopic algebras and Hamiltonians, refinement maps, renormalized states, symmetry generators, limiting local CFT, and modular-data comparison. Any additional conformal input must be explicit. Matching a finite spectrum, producing limiting commutators, or checking a finite moment problem does not supply the complete output.

The new workspace investigates a source-controlled fermion seed, a graph-norm convergence criterion, and equivariant fixed-point constructions alongside the anyonic tower. The source-grounded comparison and scoped claims are in `cft_machine/research/`.

2.3 A Quantitative Wilson-State Theorem

Theorem WT-1 [local derivation]. Assume unital C^* -algebras A_k and coherent unital injective $*$ -maps $\alpha_n^k : A_k \rightarrow A_n$ for $n \geq k$. Let ω_n be microscopic states. Suppose for each fixed k there are nonnegative $d_{k,n}$ with

$$\|\omega_{n+1} \circ \alpha_{n+1}^k - \omega_n \circ \alpha_n^k\| \leq d_{k,n}, \quad \sum_{n \geq k} d_{k,n} < \infty.$$

The norm is the dual operator norm. Set $t_k(n) = \sum_{j \geq n} d_{k,j}$. **Prove** there is a unique state ω on the inductive algebra whose restrictions are the horizontal row limits, and

$$\|\omega \circ i_k - \omega_n \circ \alpha_n^k\| \leq t_k(n).$$

- $\langle 1 \rangle 1$. Every row is norm Cauchy. BY telescoping adjacent differences and summability; for $m > n$ the difference is bounded by $\sum_{j=n}^{m-1} d_{k,j}$.
- $\langle 1 \rangle 2$. Its limit ω_k is a state with the stated tail bound. BY completeness of the dual, closure of positivity and normalization under norm limits, and $\langle 1 \rangle 1$.
- $\langle 1 \rangle 3$. The limiting family is compatible: $\omega_l \circ \alpha_l^k = \omega_k$. BY coherence, the analogous equality holds for every microscopic scale $n \geq l$; take row limits.
- $\langle 1 \rangle 4$. Define $\omega(i_k(a)) = \omega_k(a)$ on the dense algebraic union. It is well-defined, bounded, positive and unital. BY $\langle 1 \rangle 2$ – $\langle 1 \rangle 3$; continuous extension gives existence and density gives uniqueness.
- $\langle 1 \rangle 5$. The GNS maps $[a]_k \mapsto [\alpha_l^k(a)]_l$ are compatible isometries preserving cyclic vectors. BY multiplicativity, unitality, and $\omega_l(\alpha_l^k(a^*a)) = \omega_k(a^*a)$.
- $\langle 1 \rangle 6$. QED BY $\langle 1 \rangle 2$ – $\langle 1 \rangle 5$.

Effective corollary WT-2. If expectations and refinements are computable and the tail has a computable modulus, an observable $a \in A_k$ with a supplied computable bound $B \geq \|a\|$ can be evaluated in the limit to error η : choose n with $t_k(n)B < \eta/2$ and compute its row expectation to error $\eta/2$. The modulus is substantive analytic input; finitely sampled decay does not provide it.

2.4 The Generator Contract and Its Boundary

Conditional local theorem VR-1. The companion proof in `cft_machine/research/vir asoro_routes.md` constructs limiting modes and all their finite products under four explicit hypotheses: compatible isometric refinements with a positive auxiliary control operator, uniform bounds between successive graph norms, summable refinement defects in those norms, and vanishing weighted closure residuals. Its proof uses complete graph spaces and a direct telescoping identity for products.

The finite identity embeds as a projection P_N , with $P_N \rightarrow 1$ on the common smooth domain. A corner embedding is not identified with a unital observable map. Both central coefficients of CONVENTIONS (s) remain explicit. Adjoint pairing on the domain does not prove essential self-adjointness.

These two contracts must use the *same* state and representation system. WT-1 alone constructs no dynamics; VR-1 alone constructs no local OAR algebras. Positive energy, integration of smeared

modes, local-net axioms, rationality, and modular fidelity need separate proofs or carefully verified applications of existing theorems.

Numerical scope. The fermion benchmark separately checks a generator-refinement square and the pullbacks of finite Dirac ground-state covariances. Its compatible limiting chiral sea is not substituted for these nontrivial Wilson rows. The corresponding source and run paths are listed in INDEX.md and the benchmark README.

3 Wilson’s triangle, effective states, and symmetry reduction

3.1 The two directions of the construction

At fixed coarse scale k , the Wilson row consists of actual microscopic states pulled back from finer systems:

$$\omega_{k,n} = \omega_n \circ \alpha_n^k, \quad n \geq k.$$

The horizontal limit $n \rightarrow \infty$ defines ω_k ; compatibility of these limits then defines a state on the observable inductive limit. The diagram separates state convergence from merely comparing operators at two scales:

$$\begin{array}{ccccccc}
 & & & \omega_{k+2,k+2} & \longrightarrow \cdots & \omega_{k+2} & \\
 & & & \downarrow & & \downarrow & \\
 & \omega_{k+1,k+1} & \longrightarrow & \omega_{k+1,k+2} & \longrightarrow \cdots & \omega_{k+1} & \\
 & & \downarrow & \downarrow & & \downarrow & \\
 \omega_{k,k} & \longrightarrow & \omega_{k,k+1} & \longrightarrow & \omega_{k,k+2} & \longrightarrow \cdots & \omega_k
 \end{array}$$

Each horizontal row consists of functionals on its fixed coarse algebra. Vertical arrows are pullbacks by the appropriate refinement map. The rightmost column is the compatible limiting family.

WT-1 above gives norm convergence with explicit tails. It is stronger than the existence of an unspecified subsequential scaling limit. WT-2 turns the tail into an algorithm for finite-observable expectations only when a computable modulus and a computable upper bound on the observable norm are supplied. These are analytic inputs to be produced by a concrete backend.

3.2 A finite Gram consequence and a necessary warning

For $a_1, \dots, a_p \in A_k$, form $G_n(i, j) = \omega_{k,n}(a_i^* a_j)$ and $G(i, j) = \omega_k(a_i^* a_j)$. WT-3 gives

$$\|G_n - G\| \leq t_k(n) \sum_i \|a_i\|^2.$$

For a coefficient vector z , the quadratic form difference is the state difference on b^*b , $b = \sum_i z_i a_i$. WT-1 bounds it by $t_k(n)\|b\|^2$; Cauchy–Schwarz bounds $\|b\|^2 \leq \|z\|_2^2 \sum_i \|a_i\|^2$. Taking the supremum proves the claim.

Gram convergence alone does not control kernels: positive matrices $\text{diag}(1, 1/n)$ converge to $\text{diag}(1, 0)$. Likewise $R\Omega = 0$ does not imply $R = 0$ on the observable orbit. In two dimensions, choose $R = |1\rangle\langle 1|$, $\Omega = |0\rangle$ and $B = |1\rangle\langle 0| + |0\rangle\langle 1|$. Then $R\Omega = 0$ but $RB\Omega = |1\rangle$. This is the explicit VR-2 counterexample.

Evidence. `cft_machine/research/wilson_triangle.md`, WT-1–WT-3;
`cft_machine/research/virasoro_routes.md`, VR-2.

3.3 Finite-group fixed points commute with a strict algebra limit

Suppose a fixed finite group G acts by automorphisms on every A_k , and all unital injective refinements are equivariant. Then

$$\varinjlim A_k^G \cong (\varinjlim A_k)^G.$$

One inclusion is immediate. For the other, approximate any invariant element of the limit by a local element and average that approximant over G . The average stays local and invariant, and its error is no larger because the finite-group conditional expectation is contractive.

This is a strict C*-algebra statement. It does not automatically transport local nets, state-dependent weak closures, twisted sectors, or extensions. Those distinctions were retained in the broader orbifold proposal.

3.4 Invariant states and their GNS representation

EQ-1 adds invariant microscopic states. Equivariance makes every Wilson row invariant, and restricting it to fixed points cannot increase its dual-norm drift. In the full limiting GNS representation,

$$U_g[a] = [\beta_g(a)], \quad P_G = \frac{1}{|G|} \sum_g U_g$$

are well-defined unitaries and the projection onto invariant vectors. The identity $P_G \pi(a) \Omega = \pi(|G|^{-1} \sum_g \beta_g(a)) \Omega$ shows

$$\overline{\pi(A^G) \Omega} = \mathcal{H}^G.$$

Thus the fixed-algebra vacuum GNS is exactly the invariant cyclic subspace.

EQ-2 proves the represented closure identity on the full Hilbert space,

$$\pi(A^G)'' = (\pi(A)'')^G.$$

For an invariant weak-limit operator, average a weakly converging net of represented algebra elements. Finite averaging is weak-operator continuous, which gives the reverse inclusion. The equality on the full Hilbert space must be distinguished from restricting to the vacuum space $\mathcal{H}^G$.

Local versions require the action and the continuum identification to preserve the relevant interval algebras. This is exactly why independent factor parities are appropriate for products of Ising nets, while an arbitrary orbifold construction still needs additional local and sector arguments.

Evidence. `cft_machine/research/equivariant_oar.md`, EQ-1/EQ-2;
`cft_machine/research/realisation_and_fidelity.md`, fixed-point lemmas;
 Fewster–Rejzner source lines 1255–1268 for represented weak closures.

4 A Supported Ising Modular-Data Compiler

4.1 Sources, Dependencies, and Scope

Status: **explicit supported construction using existing continuum theorems.** The code and structured derivations live in `cft_machine/compiler/`; the source-controlled numerical backend is `cft_machine/benchmarks/fermion/`. This is not a constructor for arbitrary categories. Dependencies: CA-81; CONVENTIONS (u), (w), (y), (z).

4.2 Exact Input and Concrete Output

In order $(1, \sigma, \psi)$ the supported seed has

$$S = \frac{1}{2} \begin{pmatrix} 1 & \sqrt{2} & 1 \\ \sqrt{2} & 0 & -\sqrt{2} \\ 1 & -\sqrt{2} & 1 \end{pmatrix}, \quad \Theta = \text{diag}(1, e^{\pi i/8}, -1), \quad c = \frac{1}{2}.$$

The compiler compares S , twists, fusion multiplicities and the selected real c exactly, allowing unit-preserving relabelling. Rational polynomial arithmetic modulo $z^8 + 1$ represents $z = e^{\pi i/8}$ and $\sqrt{2} = z^2 - z^6$. Character T retains its separate vacuum phase $e^{-2\pi i c/24}$; it is not coerced into this smaller cyclotomic field. Explicit factor lists declare products; no general factorization algorithm or reconstruction of arbitrary F/R data is claimed.

At even size M , take antiperiodic momenta $r = -M/2 + 1/2, \dots, M/2 - 1/2$ and $\epsilon_M = 2\pi/M$. The finite microscopic mode matrix is the source-modified weighted shift

$$A_{M,n} e_r = \cos^2(\epsilon_M n/4) \frac{\sin(\epsilon_M(r - n/2))}{\epsilon_M} e_{r-n}$$

when both labels are in the window, and zero otherwise. The normal-ordered self-dual quadratic $Q_M(A_{M,n})$ includes the factor $1/2$ and the vacuum scalar; the recipe emits its actual CAR terms. The $n = 0$ sine Hamiltonian is nearest-neighbour in the real spatial basis, with an antiperiodic seam. Its positive-momentum occupation energies are positive, so the declared sea is its finite ground state. This chiral Hamiltonian is distinct from the auxiliary two-component Dirac benchmark.

The algebra map refines CAR generators by the isometric zero-padding map, $\Psi(f) \mapsto \Psi(Jf)$. It is unital and preserves conjugation and parity. A one-particle corner matrix JAJ^* is not substituted for this CAR algebra map.

4.3 The Local Observable Theory

The selected finite observable algebra is the even self-dual CAR algebra. For k factors take independent parity fixed points $(\mathbb{Z}_2)^k$. Total parity alone would retain odd-times-odd fields and give a different observable extension. Refinements are equivariant and the sea states are compatible. Thus these initial chiral states give constant Wilson rows; CA-81's nonconstant Dirac rows are a separate example of the same formalism.

The local reconstruction uses norm limits of even products of smoothly sampled, interval-supported real antiperiodic test sections. Its represented weak closure is the output interval algebra. This prescription defines the sampled-generator local net; it does not assert equality with a maximal construction using every arbitrary bounded scaling sequence.

Local convergence argument [structured derivation]. Assume a smooth real test section f supported inside a fixed interval chart, sampled as $v_M(x) = \sqrt{\epsilon_M} f(x)$. **Prove** its sharp Fourier embedding converges to f in $L^2(dx)$.

- $\langle 1 \rangle 1$. Every fixed Fourier coefficient converges. BY the Riemann sum for the coefficient in the orthonormal basis $e^{irx}/\sqrt{2\pi}$.
- $\langle 1 \rangle 2$. The norms converge to $\|f\|_2$. BY unitarity of the discrete Fourier transform and the Riemann sum $\epsilon_M \sum_x |f(x)|^2 \rightarrow \int |f|^2$.
- $\langle 1 \rangle 3$. Weak convergence and convergence of norms give strong convergence. BY density of finite Fourier sums, $\langle 1 \rangle 1$, boundedness from $\langle 1 \rangle 2$, and expansion of the squared Hilbert-space difference.
- $\langle 1 \rangle 4$. QED BY $\langle 1 \rangle 3$.

CAR norm continuity then gives convergence of every fixed even field product. Density of supported smooth tests identifies the generating local even CAR algebra. The NS vacuum representation and

the sourced even-net identity identify its vacuum interval net with $\text{Vir}_{1/2}$. The universal global circle observable algebra is not identified with a plain even CAR algebra; the source explicitly distinguishes them.

4.4 Symmetry and Modular Fidelity

The source's modified-KS theorem supplies the strong finite-energy limit and convergence of its unitary implementers in the Majorana representation. The finite Schwinger computation separately checks the self-dual central coefficient $(n^3 - n)/24$; it is not a central term of a finite matrix commutator without normal ordering.

The complete-net sector theorem at minimal-model parameter $m = 3$ gives three sectors, with weights $0, 1/16, 1/2$, Ising fusion and the displayed twists and S . This supplies fidelity of the identified local net rather than guessing its sectors from finite spectra.

For products, the compiler retains each factor's stress family and their placed sum. The sum alone is not the full tensor-product net. The sourced tensor-sector completeness result and factorwise braiding give product modular data and additive central charge. The output explicitly records the relation between input labels and canonical product labels.

This construction establishes a supported branch. General category inputs, other Ising chiralities, arbitrary central-charge lifts, and general tensor functors remain outside its implemented domain. The broader machine must expand that domain with additional constructions and proofs.

5 The Ising audit and the finite fermion evidence

5.1 A stronger local limit identification

The initial proof reconstructed the Ising interval net from sampled smooth even field products. A second skeptical audit established an upper inclusion for every uniformly bounded local lattice sequence with a represented weak limit, under the same sharp CAR refinements. This removes a possible concern that the construction selected only a convenient subset of local limits.

Lemma A2 [Lamport proof]. Assume B_M are even lattice observables supported in an interval I , with $\sup_M \|B_M\| < \infty$, and their represented images converge weakly to B on the NS even-vacuum space. **Prove** $B \in \mathcal{A}(I)$.

- $\langle 1 \rangle 1$. Let D be an even smooth field polynomial supported compactly in the complementary interval I' . Its sampled approximants D_M converge in norm by LL-1/LL-2.
- $\langle 1 \rangle 2$. Finite disjoint spatial support gives $[B_M, D_M] = 0$. Uniform boundedness and norm convergence of D_M imply $[B_M, D] \rightarrow 0$ in norm.
- $\langle 1 \rangle 3$. Weak convergence of B_M then gives $[B, D] = 0$. The complementary sampled polynomials generate $\mathcal{A}(I')$, so $B \in \mathcal{A}(I)'$.
- $\langle 1 \rangle 4$. Haag duality of the identified Ising vacuum net gives $\mathcal{A}(I)' = \mathcal{A}(I)$. QED by $\langle 1 \rangle 1$ – $\langle 1 \rangle 3$ and the sourced duality theorem.

The sampled generators supply the opposite inclusion. Thus the von Neumann algebra generated by all such bounded local limits is the same interval algebra. This statement does not make arbitrary weak convergence multiplicative, or replace the strict OAR algebra maps by a sequence quotient.

Evidence. `cft_machine/reviews/second_pass_ising.md`, A1/A2;
`cft_machine/reviews/oar_second_pass_referee.md`; KL source lines 403–407 (Haag duality); OS source lines 1451–1558 (operator convergence).

5.2 What happens to the ultraviolet doubling

The local sine Hamiltonian has additional finite low-energy excitations near the edge of the Brillouin zone. For $r_M = M/2 - 1/2$, its positive one-particle energy is

$$\frac{\sin(\epsilon_M r_M)}{\epsilon_M} = \frac{M}{2\pi} \sin(\pi/M) \longrightarrow \frac{1}{2}.$$

An even pair at the top two momenta similarly has limiting energy 2. These are not removed simply by passing to even observables.

However, the chosen sharp refinement preserves a fixed momentum label. The labels of these edge excitations keep changing with M ; their embedded vectors have no norm limit. The proved result is therefore the specified chiral OAR limit, not convergence of the entire sequence of low-energy spectra. The source explicitly permits sharp, spatially nonlocal scaling maps. Its separate wavelet-locality proposition was not misattributed to this sharp branch.

5.3 Why the modified KS prescription matters

For the complex chiral sea, the Schwinger cocycle is

$$c_S(A, B) = \text{Tr}\{SA(1 - S)BS - SB(1 - S)AS\}.$$

With the continuum shifts $B_n e_r = (r - n/2)e_{r-n}$ it reduces, for $n > 0$, to a finite sum over the n crossings of the Fermi surface:

$$\sum_{j=0}^{n-1} \left(j + \frac{1-n}{2}\right)^2 = \frac{n(n^2 - 1)}{12}.$$

The self-dual Majorana value is half of this. These are normal-ordering terms in second quantization, not central terms of bare finite matrices.

M	Complex cocycle, $n = 2$	Complex cocycle, $n = 3$
8	0.3459321593	0.7748288164
16	0.4567488187	1.5926816948
32	0.4888651513	1.8903265978
Exact continuum	1/2	2

Table 2: Source-modified finite KS modes. Exact target values are derived from the crossing sum, not fitted to the samples.

Periodic momentum wrap adds an ultraviolet crossing and cancels the cocycle in this diagnostic. The source’s characteristic-function correction removes that spurious wrap; a deliberate mutation of the implementation is rejected.

On a fixed mode core the scalar coefficient error has the analytic bound

$$\left|a_n^M(r) - (r - n/2)\right| \leq \epsilon_M^2 \left(\frac{|r - n/2|^3}{6} + \frac{n^2|r - n/2|}{16}\right).$$

Adding the bounds at adjacent scales gives a summable refinement defect. The run explicitly marks $M = 8, n = 3$ as outside its chosen shifted-core hypothesis; a displayed bound is not silently applied beyond its domain.

5.4 Actual finite Dirac states, rather than a substituted sea

The two-component Dirac benchmark has finite negative-energy projections $P_Q(r) = (1 - h_Q(r)/E_Q(r))/2$. Their row pullbacks change with Q . For fixed coarse momentum set Γ_M ,

$$\|P_{M,Q} - P_{M,\infty}\| = \sin\left(\frac{\epsilon_Q \max_{r \in \Gamma_M} |r|}{4}\right).$$

The proof then converts covariance convergence into a genuine state-norm bound. Each retained two-mode block is a pure occupied-orbital state; its density-matrix distance is twice the projection norm. The product state is handled by telescoping tensor factors, giving

$$\|\omega_{M,Q} - \omega_{M,\infty}\| \leq \frac{\epsilon_Q}{2} \sum_{r \in \Gamma_M} |r|.$$

Adjacent fine-scale drift is bounded by half this coefficient. The dyadic tail is computable and summable for each fixed row. It need not be uniform in the coarse row, and may be a loose bound at small cutoffs.

Evidence. `cft_machine/benchmarks/fermion/PROOF.md`;
`cft_machine/benchmarks/fermion/WILSON_PROOF.md`;
`cft_machine/benchmarks/fermion/runs/2026-09-22/results.toml`. The source convergence theorem supplies the analytic continuum result; the runs verify the implemented formulas and independent finite oracles.

6 A general criterion for limits of generator products

The explicit coset construction eventually gives exact finite-core equality. For other categorical models, exact stabilization may fail. VR-1 supplies a quantitative replacement: jointly construct refinements and modes with uniform graph bounds and summable defects. It is a conditional theorem, not a result already verified for arbitrary anyon chains.

6.1 Hypotheses and domain

Let $\mathcal{H}_N$ be finite-dimensional, J_N isometric, and let positive auxiliary controls $K_N \geq 1$ satisfy $K_{N+1}J_N = J_NK_N$. These controls need not be physical Hamiltonians. Set $\|v\|_{N,s} = \|K_N^s v\|$ for nonnegative integers s .

For all modes n and required orders s , assume uniform bounds

$$\|T_{N,n}v\|_{N,s} \leq C_{n,s}\|v\|_{N,s+1},$$

adjoint pairing $T_{N,n}^* = T_{N,-n}$, and summable refinement errors

$$\|(T_{N+1,n}J_N - J_NT_{N,n})v\|_{N+1,s} \leq e_{N,n,s}\|v\|_{N,s+1}, \quad \sum_N e_{N,n,s} < \infty.$$

Finally assume the weighted closure residuals tend to zero as maps between the appropriate graph spaces. Both central constants from the lab book remain explicit:

$$[T_{N,m}, T_{N,n}] - (m-n)T_{N,m+n} - \frac{\delta_{m+n,0}}{12}(m^3c^* - mc)1 = R_{N,mn}.$$

6.2 Proof mechanism VR-1

- ⟨1⟩1. Complete the algebraic inductive union in each graph norm. Exact K intertwining makes the norms consistent. Successive orthogonal finite-dimensional complements reduce K , so the completions inject into one another and the stage projections P_N are contractive and converge strongly in each graph norm.
- ⟨1⟩2. Telescope the one-step errors. On a vector from stage N , the tail norm is at most $d_{N,n,s}\|v\|_{N,s+1}$, where $d_{N,n,s} = \sum_{j \geq N} e_{j,n,s}$. Completeness constructs a limiting operator T_n ; the uniform bound extends it between graph spaces.
- ⟨1⟩3. For a word $w = (n_1, \dots, n_r)$, replace one factor at a time by its refined version. The one-step error constant is

$$E_{N,w,s} = \sum_{a=1}^r e_{N,n_a,s+a-1} \prod_{j \neq a} C_{n_j,s+j-1}.$$

It is summable. A direct fine-prefix/defect/coarse-suffix identity gives the bound without assuming $J_N J_N^* = 1$.

- ⟨1⟩4. Uniform bounds and strong convergence between successive graph spaces identify the limiting word with the product $T_{n_1} \cdots T_{n_r}$. The smooth domain given by the intersection of all graph spaces is invariant.
- ⟨1⟩5. Under the embedding, a finite identity becomes P_N , not the identity on the completed space. Since $P_N \rightarrow 1$, the residual and product limits yield the stated bracket on the smooth domain. Adjoint pairing passes by inner-product limits.
- ⟨1⟩6. QED by ⟨1⟩1–⟨1⟩5.

6.3 What must still be proved for a new backend

Finite observations of decreasing error do not prove the all-scale majorants. A useful certificate would give, for example, $e_{N,n,s} \leq a_{n,s} \rho^N$ with a proved $\rho < 1$; then its tail is explicit. The theorem's exact auxiliary-control intertwining is stringent, and no transfer to a different graph scale is implicit.

The theorem proves neither positive energy nor essential self-adjointness of real smeared fields. It does not integrate the Lie algebra, construct local observable algebras, or identify modular data. Moreover, arbitrary Hilbert isometries do not automatically come from unital OAR observable maps. The state and representation system must be the same one used by the microscopic model, not a convenient comparison Hilbert space.

Evidence. `cft_machine/research/virasoro_routes.md`, VR-1;
`cft_machine/reviews/fidelity_referee.md`. The proof was independently checked for graph-space injectivity, product-tail indexing, finite projection limits and the distinction between adjoint pairing and closed adjoints.

7 Coset Lattice Stress Generators with Exact Core Stabilization

7.1 Sources, Dependencies, and Scope

Status: source-based recipe with a local stabilization proof and exact finite witnesses. This constructs stress generators inside an ambient fermion theory. It does not yet construct the local coset lattice theory itself. Dependencies: CA-81, CA-82; CONVENTIONS (z).

7.2 An Explicit Microscopic Polynomial

Take $k + 1$ two-flavor complex fermion copies, $k \geq 1$, with half-integer momenta in the finite window and the filled negative sea. Let $t^a = \sigma_a/2$ and use the no-wrap one-particle shifts

$$j_{M,p}^a(e_r \otimes v) = e_{r-p} \otimes t^a v$$

when both labels are in the window, and zero otherwise. Normal-ordered complex second quantization gives finite quadratic currents $J_{M,p}^{u,a}$. Let I sum the first k copies, J denote the last, and $D = I + J$. For a level- l family X define

$$S_{M,n,R}^{[l]}(X) = \frac{1}{l+2} \sum_{a=1}^3 \sum_{\substack{|p| \leq R \\ |n-p| \leq R}} :X_{M,n-p}^a X_{M,p}^a:,$$

$$C_{M,n,R} = S^{[k]}(I) + S^{[1]}(J) - S^{[k+1]}(D).$$

The normal ordering orders current modes, keeping their contraction terms when the displayed products are expanded. Thus C is an explicit CAR polynomial of degree at most four, not an unspecified continuum operator.

The infinite current cocycle has coefficient $p\delta_{ab}/2$ at level one: exactly p Fermi crossings contribute $\text{Tr}(t^a t^b) = \delta_{ab}/2$. Independent copies add levels. The sourced GKO result gives

$$c = \frac{3k}{k+2} + 1 - \frac{3(k+1)}{k+3} = 1 - \frac{6}{(k+2)(k+3)}.$$

This fixes the normalization before any finite computation.

7.3 Exact Stabilization on Free-Energy Cores

Theorem CR-2 [local derivation]. Assume the no-wrap finite currents above, compatible sea GNS embeddings, and auxiliary excitation energy

$$E_M = \sum_{u,f,r>0} r n_{u,f,r} + \sum_{u,f,r<0} |r|(1 - n_{u,f,r}).$$

Let P_e project onto auxiliary energy at most e . **Prove** the embedded $C_{M,n,R}$ agrees with the continuum coset mode C_n on P_e whenever

$$R > e + |n|, \quad M/2 - 1/2 > e + 2R.$$

- $\langle 1 \rangle 1$. $[E_M, J_{M,p}] = -p J_{M,p}$. BY each nonzero bilinear moving occupation from r to $r - p$; the scalar subtraction has $p = 0$. Consequently $J_{M,p} P_e = 0$ for $p > e$ by nonnegativity of E_M . The same holds in the infinite sea.
- $\langle 1 \rangle 2$. Every omitted normally ordered current pair annihilates P_e . BY its indices summing to n : if either absolute value exceeds R , one index is positive and greater than e . Normal ordering places that mode on the right; use $\langle 1 \rangle 1$.
- $\langle 1 \rangle 3$. Retained products agree with continuum products in the safe window. BY deviations from the sea lying within $|r| \leq e$, one current action involving only $|r| \leq e + |p|$, and two retained actions involving only $|r| \leq e + 2R$.
- $\langle 1 \rangle 4$. The finite retained sum equals the continuum sum on P_e . QED BY $\langle 1 \rangle 2$ – $\langle 1 \rangle 3$.

The choice $R_M = \lfloor M/8 \rfloor$ eventually meets both conditions for every fixed e, n . A fixed word of modes uses the enlarged core $e + \sum_j |n_j|$ and therefore also stabilizes exactly. The symmetric two-index cutoff preserves finite adjoints; same-sign pairs must be grouped before taking adjoints, and zero-index pairs use the exact same-component commutation of J_0^a with J_p^a .

Vacuum corollary CR-3. For $n \geq 1$, the smaller conditions $R \geq n$ and $M/2 - 1/2 > n$ suffice for $C_{M,-n,R} \Omega = C_{-n} \Omega$. Indeed, positive and zero currents annihilate the sea, so only two negative indices of total magnitude n contribute; all involved orbitals have $|r| \leq n$.

7.4 Exact Witnesses and the Remaining Interface

The sparse-CAR producer evaluates the actual polynomials with exact complex rational amplitudes and canonical fermion signs. No many-body matrix is allocated. At $M = 8, 16$ and $R = 3, 4$, it gives

k	c	$\ C_{-2}\Omega\ ^2$	$\ C_{-3}\Omega\ ^2$
1	1/2	1/4	1
2	7/10	7/20	7/5

The independent oracle is $c(n^3 - n)/12$, not a fitted central charge. The run distinguishes the general CR-2 window from the smaller CR-3 vacuum window. CAR signs, level addition, adjoint pairings and refined descendants are checked separately; coefficient and embedding-sign mutations are rejected.

What remains is a coherent *local coset observable* construction. The ambient fermion tower retains additional sectors. A stress difference does not remove them, and finite commutants or vacuum-cyclic compressions have not been shown to commute with refinement. Global finite-scale positivity and microscopic ground states of the quartic Hamiltonian are also unproved. The result solves a generator and cutoff-control problem; it does not yet enlarge the completed modular-data compiler beyond CA-82.

8 Coset Positivity and a Scalar-Augmented OAR Tower

8.1 Sources, Dependencies, and Scope

Status: local derivations and exact finite witnesses. Dependencies: CA-83, CH-1–CH-3 in `cft_machine/research/coset_hamiltonian.md`, CO-1–CO-3 in `cft_machine/research/coset_observable_system.md`; CONVENTIONS (aa).

8.2 The Raw Finite Zero Mode Is Not Positive

For $k = 1$, fill flavor 1 at every momentum in both copies and leave flavor 2 empty. Let v be this normalized polarized occupation vector. For even M and $0 \leq R < M$, the raw coset polynomial of CA-83 obeys

$$\langle v, C_{M,0,R}v \rangle = -\frac{(M-R)(M-R-1)}{12}.$$

Proof CH-1 [structured calculation].

- $\langle 1 \rangle 1$. In each copy, $\sum_a \|J_0^a v\|^2 = M^2/4 + M/2$: the third current has eigenvalue $M/2$, and each of the other two flips one of M flavors with amplitude magnitude $1/2$. The two-copy diagonal current has squared norm $M^2 + M$.
- $\langle 1 \rangle 2$. The zero-mode Sugawara difference is $2(M^2/4 + M/2)/3 - (M^2 + M)/4 = -M(M-1)/12$.
- $\langle 1 \rangle 3$. For $1 \leq p \leq R$, each copy has $M-p$ retained flavor-flip transitions. Counting the two normally ordered terms gives a coset contribution $(M-p)/6$.
- $\langle 1 \rangle 4$. Sum $\langle 1 \rangle 3$ and add $\langle 1 \rangle 2$. QED by finite summation.

Thus the sea, whose raw energy is zero, is not a ground state when $R < M-1$. No diagonalization is needed. Exact checks give

M	$R = 3$	$R = 4$
8	-5/3	-1
16	-13	-11

The witness has auxiliary energy $M^2/2$, so it does not contradict stabilization on a fixed low-energy subspace as M grows.

8.3 Computable Vacuum-Descendant Spaces

Construction CH-2. Let V_b be the span of the vacuum and words $C_{-n_1} \cdots C_{-n_t} \Omega$ with $n_j \geq 1$ and $\sum_j n_j \leq b$. Construct them using the finite CAR polynomials at

$$R_b = 2b + 1, \quad M_b = 16(b + 1).$$

There are finitely many words. At every intermediate grade at most b , CR-2's conditions hold: $R_b > 2b$ and $M_b/2 - 1/2 = 8b + 15/2 > b + 2R_b$. Thus the finite vectors agree exactly with their continuum representatives.

Exact Gram elimination removes linear dependencies. If the retained column vectors form U_b and $G_b = U_b^* U_b$, then $P_b = U_b G_b^{-1} U_b^*$ is their orthogonal projector. Refinement identifies the same words across scales, so the isometries $J_{Bb} : V_b \rightarrow V_B$ compose exactly. On these spaces both E and C_0 act by total grade. The grade-zero space is the vacuum line.

The implemented preview through grade three uses $(\Omega, C_{-2}\Omega, C_{-3}\Omega)$ when present, with exact Gram diagonal $(1, c/2, 2c)$ and grades $(0, 2, 3)$. It uses the smaller CR-3 vacuum windows at $M = 8, 16$, not a large computation of the general schedule.

Positive Hamiltonians CH-3. The ambient operator

$$H_b = P_b E P_b + (b + 1)(1 - P_b)$$

is positive and has unique vacuum ground state: its first block has nonnegative grades and one zero vector, while its other block is strictly positive. This is a different Hamiltonian from the raw quartic zero mode. Its global projection currently has no proved local fusion-space realization.

8.4 An Independent Scalar Gives Genuine Unital OAR

Define $A_b = B(V_b) \oplus \mathbb{C}$ and, for $B \geq b$,

$$\alpha_{Bb}(a, \lambda) = (J_{Bb} a J_{Bb}^* + \lambda(1 - J_{Bb} J_{Bb}^*), \lambda).$$

The scalar is independent of a ; replacing it with a state expectation would generally preserve only complete positivity, not multiplication.

Proof CO-1.

- $\langle 1 \rangle 1$. The two first-coordinate summands have orthogonal supports. Multiplication and adjoints are preserved, and $(1, 1)$ maps to $(1, 1)$.
- $\langle 1 \rangle 2$. Compressing by J_{Bb}^* recovers a ; the second coordinate recovers λ . The map is injective.
- $\langle 1 \rangle 3$. The complementary projection splits into the newly added shell and the transported previous shell. Substitution proves $\alpha_{DB} \alpha_{Bb} = \alpha_{Db}$.
- $\langle 1 \rangle 4$. The states $\omega_b(a, \lambda) = \langle \Omega, a \Omega \rangle$ are compatible. Their Wilson rows are constant. The norm inductive limit is $K(\mathcal{H}_{\text{vac}}) + \mathbb{C}1$, and the vacuum GNS is $\mathcal{H}_{\text{vac}} = \overline{\bigcup_b V_b}$, since finite-rank operators act cyclically on Ω .
- $\langle 1 \rangle 5$. QED by $\langle 1 \rangle 1$ – $\langle 1 \rangle 4$.

The finite Hamiltonian $(E|_{V_b}, b + 1)$ is positive. Its dynamics intertwine exactly: the old subspace and the new shell are energy-reducing, and the embedded coarse algebra is scalar on the new shell. The Hamiltonians themselves need not intertwine.

8.5 What the Construction Does Not Identify

This is a new constrained observable prescription. It is generally not the restriction of the original unital CAR refinement, which retains all new-mode multiplicities. The new map instead inserts the selected sea when it transports the matrix block.

Likewise the ambient penalized resolvents tend to $P(E+i)^{-1}P$, where P projects onto the limiting vacuum module. This has a kernel on the discarded ambient sectors and is not a self-adjoint resolvent on that entire ambient Hilbert space. The new vacuum GNS representation is essential. Global constraints can be intrinsic fusion constraints. Their presence is not an obstruction to locality. What remains unproved is the identification of these descendant spaces with categorical fusion spaces, together with a local realization of the Hamiltonian and refinement operations on them.

9 Smooth Coset Symmetry Limits and Local Reconstruction

9.1 Sources and Scope

Dependencies: CA-83, CA-84; SC-1/SC-2 in `cft_machine/research/smeared_coset_limits.md`; CO-2/CO-3 in `cft_machine/research/coset_observable_system.md`; CONVENTIONS (ab).

9.2 Three Cutoffs and an Ambient Operator Theorem

For a real smooth function f on the circle, set $\hat{f}_n = (2\pi)^{-1} \int f(x) e^{-inx} dx$ and

$$\|f\|_{3/2} = \sum_n |\hat{f}_n| (1 + |n|^{3/2}).$$

The microscopic quartic operator is

$$T_M(f) = \sum_{|n| \leq K_M} \hat{f}_n C_{M,n,R_M}, \quad K_M = \lfloor M/16 \rfloor, \quad R_M = \lfloor M/8 \rfloor.$$

Its lift $A_M(f)$ uses the unital CAR refinement, not a one-particle or many-body corner extension. Each finite lift is bounded self-adjoint; its norm is not assumed uniformly bounded in M .

Common-core lemma SC-1. The algebraic finite-free-energy space decomposes orthogonally into unitary highest-weight coset modules. At the lowest unused free grade, positive modes annihilate; diagonalize the Hermitian C_0 there and generate descendants. Adjoint pairing preserves orthogonal complements. Finite free-grade multiplicity makes this procedure exhaust the space. The positive norm identity $\|C_{-1}v\|^2 = 2h\|v\|^2$ gives nonnegative lowest weights.

The sourced Carpi–Weiner bound is uniform in lowest weight at fixed c :

$$\|T(f)\xi\| \leq r_c \|f\|_{3/2} \|(1 + C_0)\xi\|.$$

It therefore applies to the module sum. The self-adjoint closures and their resolvents form direct sums, and the finite-free-energy vectors are a common core. This modulewise argument does not declare the whole ambient sector sum to be a single projective vacuum representation: its rotation phases can differ.

Theorem SC-2 [structured proof]. Prove $A_M(f)$ converges to $T(f)$ on that core, in strong resolvent sense, and through its one-parameter unitaries.

⟨1⟩1. On P_e , every retained microscopic mode agrees exactly with its continuum mode once $R_M > e + K_M$ and $M/2 - 1/2 > e + 2R_M$. The displayed schedules meet both conditions eventually; $M \geq 32(e + 1)$ suffices.

⟨1⟩2. Hence the remaining error is only a continuum Fourier tail, bounded by

$$r_c \|(1 + C_0)\xi\| \sum_{|n| > K_M} |\hat{f}_n| (1 + |n|^{3/2}) \longrightarrow 0.$$

⟨1⟩3. For core vectors,

$$(A_M - i)^{-1}(T - i)\xi - \xi = (A_M - i)^{-1}(T - A_M)\xi \longrightarrow 0.$$

The resolvent norms are at most one, and $(T - i)$ maps a graph core densely into the Hilbert space. Extend the limit by density; similarly at $-i$.

⟨1⟩4. The self-adjoint strong-resolvent theorem gives strong convergence of e^{itA_M} , uniformly on compact t intervals. Finite unitary products converge by telescoping and unit norm.

⟨1⟩5. QED by ⟨1⟩1–⟨1⟩4.

This is an ambient microscopic symmetry limit. It does not by itself remove the extra current and charge sectors or prove positivity of the raw finite Hamiltonian, which CA-84 disproves.

9.3 Local Fields in the Descendant OAR Representation

In the vacuum module of CA-84, grade selection gives the exactly finite compression

$$T_b(f) = P_b T(f) P_b = \sum_{|n| \leq b} \hat{f}_n P_b C_n P_b.$$

The CAR construction computes every retained matrix element using the safe schedule of CH-2. A computable smearing supplies only finitely many requested Fourier coefficients at each stage; no large many-body exponentiation was performed in the numerical work.

The algebra element $(e^{itT_b(f)}, 1)$ is represented as identity off V_b . Its strong limit is $e^{itT(f)}$: the zero-extended finite compressions converge on the essential core, and the preceding resolvent argument applies. The complementary scalar matters; a bare corner unitary would omit identity on the remaining Hilbert space.

Let $\mathcal{B}(I)$ be the sup-norm C^* -algebra of sequences generated by these bounded unitary approximants with real smooth f supported in I . The strong-star limit map is a $*$ -homomorphism on this generated algebra. Its kernel is a closed ideal, and the quotient identifies with the C^* -algebra generated by the limiting stress unitaries. Vacuum GNS weak closures give the sourced minimal-model Virasoro net and its complete sector data.

This represented field construction is additional to the genuine strict OAR norm limit $K(\mathcal{H}_{\text{vac}}) + \mathbb{C}1$. Strong-star-null sequences are not generally norm-null, so the quotient is not relabelled as the norm-soft inductive limit of the source.

9.4 The Local Fusion-Space Question

The finite descendant matrices currently do not retain spatial intervals: finitely many Fourier moments of functions supported in any nonempty interval can be independently specified. Nor can refinement preserve every same-smearing compression exactly, since a smearing can have all low moments zero and higher moments nonzero.

These facts concern this specific encoding, not global fusion constraints in general. The next task is to identify a categorical fusion-space basis and local interactions/refinements that realize the required matrices and limits. No such identification or bounded-range implementation is claimed in the present work.

10 Implementation, computational evidence, and skeptical review

10.1 The executable record

Every numerical headline in this report has a producing script and a run bundle. Exact arithmetic is used for modular recognition, sparse CAR calculations and Gram-coordinate OAR maps. Floating

calculations are confined to the declared finite fermion matrices and analytic-error witnesses, with explicit tolerances. No full package test suite or large many-body diagonalization was run.

Standalone suite	Assertions	Main invariant
Source-modified complex fermions	548	CAR modes, cocycles, Fourier/local Hamiltonian oracles, cutoff bounds
Self-dual Majorana specialization	96	Conjugation, quadratic factor, half-central term
Actual Dirac Wilson rows	116	Negative spectral projector, pullback composition, covariance and state tails
Exact Ising modular compiler	215	Cyclotomic data, emitted recipes, factor parity, local sampling
Sparse CAR coset stress	127	Current levels, vacuum descendants, exact central invariants, refinement signs
Coset Hamiltonian and descendants	261	Raw negative expectations, exact Gram spaces, positive repaired dynamics
Scalar-augmented OAR, generic	19	Unitality, products, metric adjoints, state and dynamics covariance
Actual descendant OAR stages	212	Computed basis embeddings and compressed modes across physical cutoffs

The retained independent referee scripts add 13 modular phase/relabel checks and 18 finite-boundary/zero-current checks. Those are distinguished from repeated executions of the main suites. The earlier orientation also reran the inherited CA-74 testset; it is not counted as new code coverage.

The fermion previews use one-particle matrices of dimension at most 64. The sparse coset calculation uses up to 96 occupation bits, not a 2^{96} allocation. The first coset suite reached 96 sparse intermediate states; the polarized Hamiltonian checks reached 2744, under a hard 5000 state guard. Commands were capped at 60 seconds, with one Julia/BLAS thread.

10.2 Actual outputs, rather than just acceptance flags

The Ising producer emits four finite instances: one or two declared factors, with 8 or 16 sites per factor. The run includes the exact modular matrices, twists and character phase, sector-label transport, every quadratic mode's terms and normal-order scalar, the spatial Hamiltonian, covariance and refinement entries. It preserves the individual stress families as well as their total sum.

The coset producer records exact rational descendant norms at $c = 1/2$ and $c = 7/10$, with distinct flags for the general CR-2 and the smaller vacuum CR-3 cutoff certificates. The Hamiltonian producer records the negative raw expectations and extracted positive spaces. The scalar-OAR producer records eight low-grade stages with exact Grams, positive complementary energies, and verified algebra/state/dynamics identities.

Source hashes accompany all producers. They allow a future run to identify which implementation

generated the data, rather than relying on a worklog claim or the appearance of a table.

10.3 Mutation tests that exposed real regressions

Deliberate corruption	What detects it
Remove the no-wrap KS correction	Spurious ultraviolet crossing and wrong central term; 33 assertions failed.
Choose the opposite Dirac spectral sea	Ground-state covariance/sign oracles; 48 assertions failed.
Remove the Majorana half factor	Wrong self-dual central normalization; 12 assertions failed.
Corrupt cyclotomic polynomial reduction	Exact field, modular and recognition identities fail.
Remove the twist-matching condition	An unsupported chirality is accepted; the rejection test fails.
Change the Sugawara denominator	Wrong coset stress invariants; 16 assertions failed.
Omit the occupation-dependent Fock insertion phase	CAR intertwining fails in the auxiliary odd-insertion witness; 13 assertions failed.
Change the positive complementary penalty	Extracted positivity/energy checks fail; 32 assertions failed.
Replace the independent OAR scalar by a vacuum character	Multiplication fails for a nonmultiplicative state pair.

All corruptions were restored before final verification. A useful late referee finding was that the first scalar-character mutation did *not* fail: the chosen product pair accidentally made the vacuum functional multiplicative on that pair. The sentinel was corrected to a pair with $\omega(ab) \neq \omega(a)\omega(b)$, and the mutation then genuinely failed. The report does not use the original green run as rejected-mutation evidence.

10.4 Substantive referee corrections

Three Astra collaborators performed research and cross-refereed work they had not authored. The environment’s four-thread limit prevented an additional fresh referee thread, so roles were reassigned for independent cross-review. The parent also inspected the current sources/code and reran the main checks.

The reviews made the following material changes to the scientific record:

1. Separated a generator-refinement square from Wilson rows of actual microscopic states, and added the nontrivial Dirac-state construction.
2. Replaced an insufficient stress-subnet argument by the full even local CAR reconstruction and direct even-Fermi/Ising source identity.
3. Required independent factor parities and retained each factor stress family, avoiding both a larger observable extension and a smaller diagonal net.
4. Added the computable norm upper bound required by the effective Wilson expectation algorithm.
5. Tracked the actual central charge of the holomorphic orbifold recipe; modular matching alone does not satisfy an arbitrary prescribed lift.
6. Corrected finite coset adjointness at zero-current indices and sharpened the support threshold without increasing compute budgets.

7. Distinguished the norm limit $K(\mathcal{H}) + \mathbb{C}1$ from its represented local field limits, and the new constrained refinement from the original unital CAR refinement.
8. Established the stronger bounded-local-WOT Ising upper inclusion, while explicitly recording ultraviolet states excluded by the chosen RG.

10.5 Practical failures and what they taught

Two transient Astra capacity errors were retried. A source extraction lost the factor 2 in the GKO Sugawara coefficient; the original PDF was visually checked. Author TeX was kept unchanged, including its whitespace and hashes. Mutation logs were retained as text artifacts because ordinary LaTeX-style `.log` files are ignored by the repository.

The first integrated descendant-matrix test timed out at the fixed 60-second limit. It recomputed each sparse action once per matrix row. The corrected code computes each necessary action once and uses the proved energy grading to skip exactly vanishing columns; the operator is unchanged. The final 212-assertion integration suite passed within the original cap.

Evidence. Run bundles below `cft_machine/benchmarks/` and `cft_machine/compiler/`; referee records below `cft_machine/reviews/`; root `INDEX.md` and worklogs 014–015.

11 The fusion-space locality target and the unfinished machine

11.1 Global admissibility is not a locality obstruction

Tobias’s final direction is essential: globally constrained Hilbert spaces can simply be fusion spaces. A local Hamiltonian on such a space need not be local on an unconstrained tensor product with independent on-site labels. The correct question is which fusion labels and vertices a term acts on, how its matrix elements depend on surrounding admissible data, and how that action behaves under refinement.

Consequently, the descendant-space construction should neither be dismissed because it is globally constrained nor declared local because it is finite. Its missing interface is specific: an identification with categorical fusion paths and a local operator calculus on those paths. The current project has fusion-tree conventions and exact Fibonacci local-operator machinery, but no proved identification between those spaces and the new coset descendant spaces has been obtained.

11.2 What a local fusion-space realization would have to establish

This is a research target, not a claimed theorem. For a chosen input category and microscopic size, the next construction should specify:

1. An admissible fusion-path or fusion-tree basis, with unit, bracketing, charge and inner-product conventions fixed before importing matrices.
2. A map from the constructed microscopic state space into that fusion space, with a proved inverse or explicitly controlled image.
3. Hamiltonian densities and symmetry ingredients acting on a bounded neighbourhood of fusion data, including any required F-moves.
4. Refinement maps compatible with those admissibility constraints, the selected states and the local observable interpretation.
5. The existing OAR and microscopic-symmetry convergence hypotheses in that realization, plus the sector comparison establishing modular fidelity.

The global constraint should be carried by the space itself. A generic full-chain penalty projector is not a substitute for the local operator description. Conversely, the existence of such a projector is not a proof that a local fusion-space description is impossible.

11.3 Status of the candidate routes

Route	Established here	Remaining interface
Ising and declared products	Exact modular-input compiler, local Majorana Hamiltonians, strict CAR OAR, local interval net and sector identification	Other chiralities/lifts and general input categories; no generic F/R solver.
Coset current construction	Explicit quartic modes; exact finite-core stabilization; smooth resolvent and unitary limits	Local positive fusion-space Hamiltonian and observable realization without ambient sectors.
Extracted descendant OAR	Computable spaces, positive Hamiltonians, unital scalar-augmented maps, compatible dynamics, minimal-net reconstruction	Categorical fusion-space identification and local action/refinement.
Finite-group orbifolds	Sourced continuum existence families and proved fixed-point transport lemmas	Microscopic seeds, local extensions, twisted sectors and local convergence.
Discrete holomorphic/Sugawara	Sourced exact lattice Virasoro in its own correlation-field quotient	Positive quantum realization and an OAR bridge; that quotient is not automatically a GNS quotient.
Anyonic variable-number tower	Existing finite categorical algebras, refinement and residual machinery	Compatible dynamics and states with proved continuum locality, symmetry and categorical fidelity.

11.4 Why the universal algorithm is still open

The positive constructions do not prove that every unitary modular category has a conformal realization with computable microscopic data. Existing source realization theorems for particular families do not automatically give lattice Hamiltonians or OAR maps. Numerical recognition of a critical spectrum would not fill that gap.

Nor is a general functor implemented. The Ising code transports declared presentation labels and factor data. A functor on a broader category of inputs must specify which tensor morphisms preserve the physical choices, how states and refinements are transported, and how compositions agree. The current research notes discuss decorated equivalences as an initial domain, without claiming arbitrary tensor functors preserve criticality.

The proposed tricritical-Ising modular recognizer was stopped when wind-up was requested. Its six-sector formula proposal is recorded as unimplemented; there is no tricritical input compiler or associated acceptance test to claim. The $c = 7/10$ results in this report concern actual coset stress and descendant constructions, not an already completed new modular-input frontend.

11.5 What is retained for the next investigation

The useful output is not merely a list of gaps. The repository now contains actual microscopic formulas, exact finite vectors and Gram matrices, state and operator convergence criteria, controlled

symmetry limits, positive finite constrained models, and reproducible regression witnesses. These can be tested against a candidate local fusion-space model instead of rebuilding the continuum machinery from scratch.

The work is wound up at the present verified boundary. No general category-to-CFT result is claimed, and no pending proposal is represented as implemented. The next scientific step is the local fusion-space identification specified above.

A Source map and reproduction commands

A.1 Primary mathematical sources

All external mathematical claims use local source files. The corresponding manifests retain bibliography, legal retrieval URLs, retrieval date, hashes and extraction details. The following map identifies the sources carrying the central claims in this report.

OS: Osborne–Stottmeister.

Conformal field theory from lattice fermions, arXiv:2107.13834. Local extraction: `references/text/CFTFromLatticeFermions.txt`. Theorem A and the full mode-convergence theorem; Definitions 3.1 and 3.9; normal-ordering/cocycle formulas; modified KS equations (197)–(199); strong-resolvent, unitary and correlation limits. Exact ranges used appear in the individual proof notes and report source comments.

Wavelet OAR framework.

arXiv:2010.11121, local file `literature/md/2010.11121/2010.11121.md`. Lines 101–111 and 126–180 define the strict observable maps, Wilson rows, compatible limiting states and GNS representations. These definitions are the baseline for deciding whether a proposed construction is actually OAR.

Soft inductive-limit framework.

arXiv:2306.16063, local file `literature/md/2306.16063/2306.16063.md`. Lines 185–208 and 568–593 provide the contractive-map and norm-null definitions. The distinction from a strong-star-null quotient is explicit.

GKO: Goddard–Kent–Olive.

Unitary representations of the Virasoro and super-Virasoro algebras, Communications in Mathematical Physics 103 (1986), 105–119, DOI 10.1007/BF01464283. Original and extraction: `references/cft-machine/virasoro/GoddardKentOlive1986/`. Printed pages 107–108 were visually checked because the extraction loses a coefficient factor. Equations (2.5)–(2.20) fix the current/Sugawara and diagonal-coset normalization.

CW: Carpi–Weiner.

On the uniqueness of diffeomorphism symmetry in Conformal Field Theory. arXiv:math/0407190; Communications in Mathematical Physics 258 (2005), 203–221, DOI 10.1007/s00220-005-1335-4. Local source: `references/cft-machine/coset-symmetry/CarpiWeiner2004/source.tex`. Lines 721–788 give the mode-energy and Fourier bounds; 864–976 provide essential self-adjointness and resolvent/unitary control.

KL: Kawahigashi–Longo.

Classification of Local Conformal Nets. Case $c < 1$, arXiv:math-ph/0201015. Local source: `references/cft-machine/realisation/KawahigashiLongo2002/source.tex`. Lines 403–407 give Haag duality; 647–669 give minimal fusion and weights; 1008–1024 identify the diagonal coset; 1027–1157 give complete rationality, all sectors, twists and modular data.

KLM: Kawahigashi–Longo–Müger.

Multi-interval Subfactors and Modularity of Representations in Conformal Field Theory, arXiv:math/9903104. Local source:

references/cft-machine/realisation/KawahigashiLongoMuger1999/source.tex. Lines 969–977 and 1518–1551 supply the hypotheses and completeness of tensor-product sectors used by the product Ising branch.

BO and LMR: the Ising observable bridge.

Böcken-hauer, *Localized Endomorphisms of the Chiral Ising Model*, arXiv:hep-th/9407079; Longo–Martinetti–Rehren, *Geometric modular action for disjoint intervals and boundary conformal field theory*, arXiv:0912.1106. Sources under references/cft-machine/realisation/. BO lines 324–355 and 563–588 distinguish local even CAR, the NS vacuum space and the universal-circle caveat. LMR lines 1165–1168 directly identify the real-Fermi even subnet with $\text{Vir}_{1/2}$.

RSW: Rowell–Stong–Wang.

On classification of modular tensor categories, arXiv:0712.1377. Local source: references/category-theory/RowellStongWang2009Classification/source/RSWfinal3.tex. Lines 689–690 give balancing; 2441–2474 give the exact Ising modular datum.

Realization alternatives and fidelity counterexamples.

Evans–Gannon, arXiv:1804.11145; Bischoff, arXiv:1803.04949; Mignard–Schauenburg, arXiv:1708.02796. Local sources and exact anchors are registered in references/cft-machine/realisation/SOURCES.md. Their continuum constructions and counterexamples are not silently promoted to microscopic lattice/OAR theorems.

C*-algebras, weak closures, and proof presentation.

Landsman, arXiv:math-ph/9807030, source under references/cft-machine/coset-observables/; Fewster–Rejzner source under references/qft/FewsterRejzner2019/; Lamport, *How to Write a Proof*, author source under references/cft-machine/proofs/. These support the explicit algebraic quotient/GNS statements and the structured proof notation.

A.2 Where to read the complete proofs

The report is a detailed synthesis; the checked source-linked derivations remain accessible as short repository notes:

- cft_machine/compiler/PROOF.md and cft_machine/compiler/LOCAL_LIMIT.md: exact Ising input, microscopic construction, local reconstruction and modular fidelity.
- cft_machine/research/wilson_triangle.md and cft_machine/research/equivariant_oar.md: states, effective estimates, finite-group reduction and represented closures.
- cft_machine/research/virasoro_routes.md: graph-norm products and the explicit vacuum-null counterexample.
- cft_machine/research/coset_route.md and cft_machine/research/smeared_coset_limits.md: actual quartic modes, exact finite-core stabilization, smooth resolvents and unitaries.
- cft_machine/research/coset_hamiltonian.md and cft_machine/research/coset_observable_system.md: raw negativity, positive extracted models, genuine strict OAR and local field reconstruction.
- cft_machine/reviews/: skeptical cross-referees, resolved findings, the second Ising audit, and retained independent check scripts.

A.3 Reproduce the code and artifacts

Run commands from the repository root. The test scripts use the current Julia project and mostly only standard libraries. Keep one thread and the explicit per-command cap; the complete historical `Pkg.test()` suite remains suspended by the laptop policy.

```
OPENBLAS_NUM_THREADS=1 JULIA_NUM_THREADS=1 timeout 60 \  
  julia --project=. cft_machine/benchmarks/fermion/test.jl  
OPENBLAS_NUM_THREADS=1 JULIA_NUM_THREADS=1 timeout 60 \  
  julia --project=. cft_machine/benchmarks/fermion/test_wilson.jl  
OPENBLAS_NUM_THREADS=1 JULIA_NUM_THREADS=1 timeout 60 \  
  julia --project=. cft_machine/compiler/test.jl  
OPENBLAS_NUM_THREADS=1 JULIA_NUM_THREADS=1 timeout 60 \  
  julia --project=. cft_machine/benchmarks/coset/test.jl
```

The same prefix runs the test scripts under `cft_machine/benchmarks/coset_hamiltonian/` and `cft_machine/benchmarks/unitized_oar/`; the latter has both `test.jl` and `test_descendants.jl`. Each benchmark's `run.jl` regenerates its own dated TOML bundle. The two independent scripts are `cft_machine/reviews/check_compiler.jl` and `cft_machine/reviews/check_coset.jl`.

Rebuild this synthesis with `pdflatex` twice for the contents and hyperlinks:

```
pdflatex -interaction=nonstopmode -halt-on-error \  
  -output-directory=cft_machine cft_machine/report.tex  
pdflatex -interaction=nonstopmode -halt-on-error \  
  -output-directory=cft_machine cft_machine/report.tex
```

The project-wide lab book has its own mandatory commands:

```
make check-report-shards  
make report
```

The main scientific record remains the sources, proofs, implementations, tests, manifests and run artifacts together. Reproducibility is not supplied by the PDF alone; the PDF points to the complete record.